

Class / inheritance sketch

SystemVerilog testbenches lean on classes long before UVM

Objects versus components

- On the object side, a base class carries a name
- Siblings like SpiPacket share the same parent but different extras
- On the component side, CompBase knows parent and name
- Objects are data handles you create and pass
- Extends means the child inherits every member unless it overrides a virtual method

Browser lab

Digital Design and Verification Platform

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INTERACTIVE TOOL

Class / inheritance sketch

Diagram-only literacy for SystemVerilog TB OOP: **objects** (transactions) vs **components** (hierarchy), and how extends shares fields and methods. Starter: **UartPacket** extends **Packet**.

Starter example: object tree with UartPacket selected (extends Packet). Toggle inherited members, or switch to the component tree.

Load starter example

Challenges 0 / 22 cleared

Quiz: extends: In SV, class Child extends Parent means...

☐ Child inherits Parent members and can add more ☐ Child replaces the Parent module in synthesis ☐ Child is always a wire

☐ extends only works inside always_ff

Show hint Check Next Idle

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22

Core ideas

extends Child gets parent fields/methods.	Object Transaction / data — new() handle.	Component Hierarchy member — build/run roles.	Override Child virtual method replaces parent.
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Browser lab starter

Real SV TB track practice

- In the real track, open this module's examples prompts
- Restate the two trees in one sentence, objects for data, components for hierarchy
- On paper, sketch Packet with two children
- Then sketch CompBase with Driver and Monitor as siblings and UartDriver extending Driver
- Optional: find one extends in a local testbench or UVM example and label parent and child
- No compile required, the goal is to read inheritance, not run a full class runtime

Pitfalls to watch

- Do not treat every class as a component
- Do not copy fields into every child when extends already shares them
- Do not forget virtual when you mean override
- And remember

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser
- On paper, draw one object tree and one component tree with extends arrows labeled
- When you are ready